

606020



Test Booklet No.

ENGLISH

Test Booklet Code

05

NEET RANKERS

TEST SERIES-2025



Do not open this Test Booklet until you are asked to do so.

Important Instructions:

- The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with blue/black ball point pen only.
- The test is of 3 hours 20 minutes duration and the Test Booklet contains 200 multiple-choice questions (four options with a single correct answer) from Physics, Chemistry and Biology (Botany and Zoology). 50 questions in each subject are divided into two Sections (A and B) as per details given below:
 - Section A shall consist of 35 (Thirty-five) Questions in each subject (Question Nos - 1 to 35, 51 to 85, 101 to 135 and 151 and 185). All questions are compulsory.
 - Section B shall consist of 15 (Fifteen) questions in each subject (Question Nos - 36 to 50, 86 to 100, 136 to 150, and 186 to 200). In Section B, a candidate needs to attempt any 10 (Ten) questions out of 15 (Fifteen) in each subject.

Candidates are advised to read all 15 questions in each subject of Section B before they start attempting the question paper. In the event of a candidate attempting more than ten questions, the first ten questions answered by the candidate shall be evaluated.
- Each question carries 4 marks. For each correct response, the candidate will get 4 marks. For each incorrect response, one mark will be deducted from the total scores. The maximum marks are 720.
- Use Blue/Black Ball Point Pen only for writing particulars on this page/marking responses on Answer Sheet.
- Rough work is to be done in the space provided for this purpose in the Test Booklet only.
- On completion of the test, the candidate must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
- The CODE for this Booklet is 05. Make sure that the CODE printed on the Original Copy of the Answer Sheet is the same as that on this Test Booklet. In case of discrepancy, the candidate should immediately report the matter to the Invigilator for replacement of both the Test Booklet and the Answer Sheet.
- The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Roll No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
- Use of white fluid for correction is NOT permissible on the Answer Sheet.
- Each candidate must show on-demand his/her Admit Card to the Invigilator.
- No candidate, without special permission of the centre Superintendent or Invigilator, would leave his/her seat.
- The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet twice. Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.
- Use of Electronic/Manual Calculator is prohibited.
- The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
- No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.
- The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.
- Compensatory time of one hour five minutes will be provided for the examination of three hours and 20 minutes duration, whether such candidate (having a physical limitation to write) uses the facility of Scribe or not.

Name of the Candidate (in Capitals): _____

Roll Number : In figures _____

: In words _____

Centre of Examination (in Capitals): _____

Candidate's Signature: _____ Invigilator's Signature: _____

Facsimile Signature stamp of Centre Superintendent

Physics : Section-A (Q. No. 1 to 35)

- When a body is taken from the equator to the poles, its weight;
 - remains constant
 - increases
 - decreases
 - increases at n-pole and decreases at s-pole
- Current i is carried in a wire of length L . If the wire is turned into a circular coil, the maximum magnitude of torque in a given magnetic field B will be;

(1) $\frac{L^2 B^2 i}{2}$	(2) $\frac{L^2 B i}{2}$
(3) $\frac{L^2 i B}{4\pi}$	(4) $\frac{L^2 B i}{2\pi}$
- To obtain p -type semiconductor, we need to dope pure Si with;

(1) aluminium	(2) phosphorous
(3) oxygen	(4) germanium
- A stone falls from a balloon that is descending at a uniform velocity of 12 ms^{-1} . The displacement of the stone from the point of release after 10 seconds is; ($g = 9.8 \text{ m/s}^2$)

(1) 490 m	(2) 510 m
(3) 610 m	(4) 750 m
- A thermo-dynamic system is taken from state (P_1, V_1) to (P_2, V_2) by two different process. The quantity which will remain same in both process will be;

(1) ΔQ	(2) ΔW
(3) $\Delta Q + \Delta W$	(4) $\Delta Q - \Delta W$
- In hydrogen atom, if the difference in the energy of the electron in $n = 2$ and $n = 3$ orbits is E , the ionisation energy of hydrogen atom is;

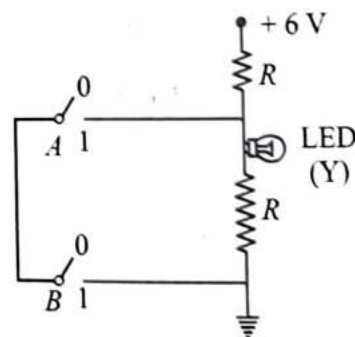
(1) $13.2E$	(2) $7.2E$
(3) $5.6E$	(4) $3.2E$

- If the amplitude ratio of two sources producing interference is 3 : 5, then the ratio of intensities at maxima and minima is;

(1) 25 : 16	(2) 16 : 1
(3) 5 : 3	(4) 25 : 9
- 220 V, 50 Hz, AC is applied to a resistor. The instantaneous value of voltage is;

(1) $220\sqrt{2} \sin 100 \pi t$
(2) $220 \sin 100 \pi t$
(3) $220\sqrt{2} \sin 50 \pi t$
(4) $220 \sin 50 \pi t$
- Given below are two statements:
Statement-I: A charge is moving in a uniform magnetic field. The charge may experience no magnetic force.
Statement-II: Kinetic energy of a charged particle moving in a uniform magnetic field may change.
 In the light of the above statements, choose the correct answer from the options given below:

(1) Statement-I is correct, but Statement-II is incorrect.
(2) Statement-I is incorrect, but Statement-II is correct.
(3) Statement-I and Statement-II both are correct.
(4) Statement-I and Statement-II both are incorrect.
- The correct Boolean operation represented by the circuit diagram drawn is;



- | | |
|---------|----------|
| (1) OR | (2) NAND |
| (3) NOR | (4) AND |

11. For hydrogen gas $C_p - C_v = a$ and for oxygen gas $C_p - C_v = b$ then the relation between a and b is;

(where C_p and C_v are specific heat capacity)

- (1) $a = 16b$ (2) $a = b$
(3) $b = 16a$ (4) $a = 4b$

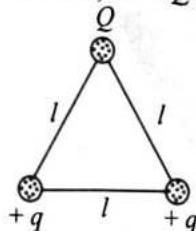
12. The electric field due to an electric dipole at a distance r from its centre in axial position is E . If the dipole is rotated through an angle of 90° about its perpendicular axis, then the magnitude of electric field at the same point will be;

- (1) E (2) $\frac{E}{4}$
(3) $\frac{E}{2}$ (4) $2E$

13. The dimensions of $\frac{a}{b}$ in the equation $p = \frac{a - t^2}{bx}$, where p is pressure, x is distance and t is time, are;

- (1) $[M^2LT^{-3}]$ (2) $[MT^{-2}]$
(3) $[LT^{-3}]$ (4) $[ML^3T^{-1}]$

14. Three charges Q , $+q$ and $+q$ are placed at the vertices of an equilateral triangle of side l as shown in the figure. If the net electrostatic energy of the system is zero, then Q is equal to;



- (1) $-\frac{q}{2}$ (2) $-q$
(3) $+q$ (4) $+\frac{q}{2}$

15. Match List-I with List-II

List-I	List-II
A. Stress $\times$ strain	I. J
B. $\frac{YA}{l}$	II. Nm^{-1}
C. Yl^3	III. Jm^{-3}
D. $\frac{Fl}{AY}$	IV. m

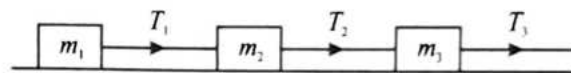
Choose the correct answer from the options given below:

- (1) A-III, B-II, C-I, D-IV
(2) A-III, B-II, C-IV, D-I
(3) A-II, B-I, C-III, D-IV
(4) A-II, B-III, C-IV, D-I

16. Drift velocity v_d varies with the intensity of electric fields as per the relation;

- (1) $v_d \propto E$ (2) $v_d \propto \frac{1}{E}$
(3) $v_d = \text{constant}$ (4) $v_d \propto E^2$

17. Three blocks of masses m_1 , m_2 and m_3 are connected by massless strings as shown on a frictionless table. They are pulled with a force $T_3 = 40$ N. If $m_1 = 10$ kg, $m_2 = 6$ kg and $m_3 = 4$ kg, the tension T_2 will be;



- (1) 20 N (2) 40 N
(3) 10 N (4) 32 N

18. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R)

Assertion (A): When wavelength of incident radiation in a photoelectric experiment is decreased, then the stopping potential will decrease.

Reason (R): With decrease in wavelength, energy of incident light increases.

In the light of the above statements, choose the correct answer from the options given below:

- (1) (A) is false but (R) is true.
(2) (A) is true but (R) is false.
(3) Both (A) and (R) are true and (R) is the correct explanation of (A).
(4) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

19. Multiply 107.88 by 0.610 and express the result with correct number of significant figures;

- (1) 65.8068 (2) 64.807
(3) 65.81 (4) 65.8

20. The potentials of the two plates of capacitor are +10 V and -10 V. The charge on one of the plates is 40 C. The capacitance of the capacitor is;
 (1) 2 F (2) 4 F
 (3) 0.5 F (4) 0.25 F
21. Two different loops are concentric and lie in the same plane. The current in the outer loop is clockwise and increasing with time. The induced current in the inner loop is;
 (1) clockwise
 (2) zero
 (3) counter-clockwise
 (4) in a direction that depends on the ratio of the loop radii
22. The wavelength of light in two liquids x and y is 3500 Å and 7000 Å respectively, then the critical angle of x relative to y will be;
 (1) 60° (2) 45°
 (3) 30° (4) 15°
23. Moment of inertia of a body depends upon;
 (1) axis of rotation
 (2) torque
 (3) angular momentum
 (4) angular velocity
24. The Young's modulus of the material of a wire is equal to the;
 (1) stress required to increase its length four times
 (2) stress required to produce unit strain
 (3) strain produced in it
 (4) half the strain produced in it
25. In Bohr's model, the atomic radius of the first orbit is r_0 , then the radius of the third orbit is;
 (1) $\frac{r_0}{9}$ (2) r_0
 (3) $9r_0$ (4) $3r_0$
26. A sphere is released on a smooth inclined plane from the top. When it moves down, its angular momentum is;
 A. Conserved about every point
 B. Conserved about the point of contact only
 C. Conserved about any point on a line parallel to the inclined plane and passing through the centre of the ball.
 Choose the *most appropriate* answer from the options given below;
 (1) A only (2) C only
 (3) B and C only (4) A, B and C
27. A uniform chain of length L and mass M is lying on a smooth table and one third of its length is hanging vertically down over the edge of the table. If g is acceleration due to gravity, the work required to pull the hanging part on the table is;
 (1) MgL (2) $\frac{MgL}{3}$
 (3) $\frac{MgL}{9}$ (4) $\frac{MgL}{18}$
28. Escape velocity from a planet is v_e . If its mass is increased to 16 times and its radius is increased to 4 times, then the new escape velocity would be;
 (1) v_e (2) $\sqrt{2}v_e$
 (3) $2v_e$ (4) $2\sqrt{2}v_e$
29. The horizontal range of a projectile is $4\sqrt{3}$ times of its maximum height. The angle of projection will be;
 (1) 60° (2) 37°
 (3) 30° (4) 45°
30. A mass of 1 kg is acted upon by a single force $F = (4\hat{i} + 4\hat{j}) N$. Under this force, it is displaced from (0, 0) to (2m, 2m). If initially the speed of the particle was 2 ms^{-1} , its final speed should be;
 (1) 6 ms^{-1} (2) 4.5 ms^{-1}
 (3) 8 ms^{-1} (4) 4 ms^{-1}

31. An electromagnetic wave going through vacuum is described by $E = E_0 \sin(kx - \omega t)$. Which of the following is independent of wavelength?

(1) k (2) ω
(3) k/ω (4) $k\omega$

32. A particle starts with a velocity of 2 ms^{-1} and moves in a straight line with a retardation of 0.1 ms^{-2} . The first time at which the particle is 15 m from the starting point is:

(1) 10 s (2) 20 s
(3) 30 s (4) 40 s

33. Velocity at mean position of a particle executing SHM is v . velocity of the particle at a distance equal to half of the amplitude will be;

(1) $\frac{v}{2}$ (2) $\frac{v}{\sqrt{2}}$
(3) $\frac{\sqrt{3}}{2}v$ (4) $\frac{\sqrt{3}}{4}v$

34. Two vessels A and B having equal volume contain equal masses of hydrogen in A and helium in B at 300 K. Then, mark the correct statement?

(1) The pressure exerted by hydrogen is half that exerted by helium
(2) The pressure exerted by hydrogen is equal to that exerted by helium
(3) Average KE of the molecules of hydrogen is half the average KE of the molecules of helium
(4) The pressure exerted by hydrogen is twice that exerted by helium

35. A body at rest breaks into two pieces of equal masses. The parts will move;

(1) in same direction
(2) along different lines
(3) in opposite directions with equal speeds
(4) in opposite directions with unequal speeds

Physics : Section-B (Q. No. 36 to 50)

36. Two equal drops of water are falling through air with a steady velocity v . If the drops coalesce, the new velocity will be;

(1) $2v$ (2) $\sqrt{2}v$
(3) $2^{2/3}v$ (4) $\frac{v}{\sqrt{2}}$

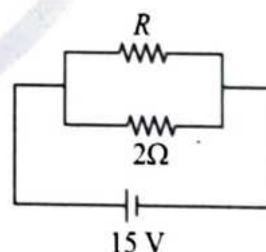
37. In Young's double slit experiment, the fringe width will remain same, if (D = distance between screen and plane of slits, d = separation between two slits and λ = wavelength of light used);

(1) Both λ and D are doubled
(2) Both d and D are doubled
(3) D is doubled but d is halved
(4) λ is doubled but d is halved

38. The centre of a wheel rolling on a plane surface moves with a speed v_0 . A particle on the rim of the wheel at the same level as the centre will be moving at speed;

(1) zero (2) v_0
(3) $\sqrt{2}v_0$ (4) $2v_0$

39. If in the circuit, power dissipation is 150 W, then R is;



(1) 2Ω (2) 6Ω
(3) 5Ω (4) 4Ω

40. The upper half of an inclined plane of inclination θ is perfectly smooth while lower half is rough. A block starting from rest at the top of the plane will again come to rest at the bottom, if the

coefficient of friction between the block and lower half of the plane is given by;

- (1) $\mu = \frac{1}{\tan \theta}$ (2) $\mu = \frac{2}{\tan \theta}$
(3) $\mu = 2 \tan \theta$ (4) $\mu = \tan \theta$

41. A body of mass m is attached to the lower end of a spring whose upper end is fixed. The spring has negligible mass. When the mass m is lightly pulled down and released, it oscillates with a time period of 3 s. When the mass m is increased by 1 kg, the time period of oscillations becomes 5 s. The value of m in kg is;

- (1) $\frac{3}{4}$ (2) $\frac{4}{3}$
(3) $\frac{16}{9}$ (4) $\frac{9}{16}$

42. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R)

Assertion (A): If a current flows through a wire of non-uniform cross-section, potential difference per unit length of wire is same throughout the length of wire.

Reason (R): Current through the wire is same at all cross-sections.

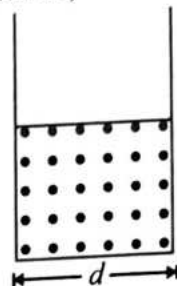
In the light of the above statements, choose the correct answer from the options given below:

- (1) (A) is false but (R) is true.
(2) (A) is true but (R) is false.
(3) Both (A) and (R) are true and (R) is the correct explanation of (A).
(4) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

43. The force per unit length between two long parallel wires A and B carrying current is 0.004 Nm^{-1} . The conductors are 0.01 m apart. If the current in conductor A is twice that of conductor B, then the current in the conductor B would be;

- (1) 5 A (2) 50 A
(3) 10 A (4) 100 A

44. A parallel plate air capacitor has a capacitance C . When it is half filled with a dielectric of dielectric constant 5, then the percentage increase in the capacitance will be;



- (1) 400% (2) 66.6%
(3) 33.3% (4) 200%

45. The image formed by a convex mirror of focal length 30 cm is a quarter of the size of the object. The distance of the object from the mirror is;

- (1) 30 cm (2) 90 cm
(3) 120 cm (4) 60 cm

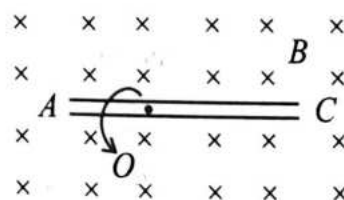
46. In an $L-C-R$ series AC circuit, at resonance

- (1) the capacitive reactance is more than the inductive reactance
(2) the capacitive reactance equals the inductive reactance
(3) the capacitive reactance is less than the inductive reactance
(4) the current is minimum

47. If the electric flux entering and leaving an enclosed surface respectively is ϕ_1 and ϕ_2 , then the electric charge inside the surface will be;

- (1) $(\phi_1 + \phi_2)\epsilon_0$ (2) $(\phi_2 - \phi_1)\epsilon_0$
(3) $\frac{\phi_1 + \phi_2}{\epsilon_0}$ (4) $\frac{\phi_2 - \phi_1}{\epsilon_0}$

48. A conducting rod AC of length $4l$ is rotated about a point O in a uniform magnetic field B directed into the paper. If $AO = l$ and $CO = 3l$, then;





$$(1) V_A - V_O = \frac{B\omega l^2}{2}$$

$$(2) V_O - V_C = \frac{7}{2} B\omega l^2$$

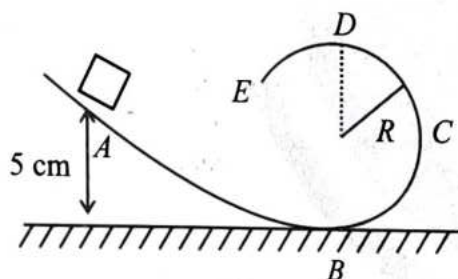
$$(3) V_A - V_C = 4B\omega l^2$$

$$(4) V_C - V_O = \frac{9}{2} B\omega l^2$$

49. Light with an average flux of 20 W/cm^2 falls on a non-reflecting surface at normal incidence having surface area 20 cm^2 . The energy received by the surface during time span of 1 min is;

- (1) $12 \times 10^3 \text{ J}$ (2) $24 \times 10^3 \text{ J}$
 (3) $48 \times 10^3 \text{ J}$ (4) $1.0 \times 10^3 \text{ J}$

50. A frictionless track ABCDE ends in a circular loop of radius R . A body slides down the track from point A which is at height $h = 5 \text{ cm}$. Maximum value of R for a body to complete the loop successfully is;



- (1) 2 cm (2) $\frac{10}{3} \text{ cm}$
 (3) $\frac{15}{4} \text{ cm}$ (4) $\frac{18}{3} \text{ cm}$

Chemistry : Section-A (Q. No. 51 to 85)

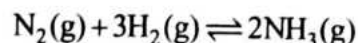
51. Which of the following would have a permanent dipole moment?

- (1) BF_3 (2) SiF_4
 (3) SF_4 (4) XeF_4

52. Radiation corresponding to the transition $n = 4$ to $n = 2$ in hydrogen atoms falls on a certain metal (work function = 2.5 eV). The maximum kinetic energy of the photo-electrons will be;

- (1) 0.5 eV (2) 2.55 eV
 (3) 4.45 eV (4) 0.05 eV

53. The direction of the given reaction is in forward direction when;



- (1) $Q < K_c$ (2) $Q = 0$
 (3) $Q = K_c$ (4) $Q > K_c$

54. Which is true about ψ ?

- (1) ψ represents the probability of finding an electron around the nucleus.
 (2) ψ represent the amplitude of the electron wave.
 (3) Both (1) and (2)
 (4) None of these

55. Given below are two statements:

Statement-I: In CrO_5 , oxidation number of Cr is +10.

Statement-II: H_3PO_2 is tribasic acid.

In the light of the above statements, choose the most appropriate answer from the options given below :

- (1) Statement-I is correct but Statement-II is incorrect.
 (2) Statement-I is incorrect but Statement-II is correct.
 (3) Both Statement-I and Statement-II are correct.
 (4) Both Statement-I and Statement-II are incorrect.

56. Warming ammonium chloride with sodium hydroxide in a test tube is an example of;

- (1) Closed system
 (2) Isolated system
 (3) Open system
 (4) Cyclic system

57. Percentage of Se in peroxidase anhydrase enzyme is 0.25% by weight (at. weight = 78.4 u), then minimum molecular weight of peroxidase anhydrase enzyme is;

- (1) $3.136 \times 10^3 \text{ g mol}^{-1}$
- (2) 31.36 g mol^{-1}
- (3) 3136 g mol^{-1}
- (4) $3.136 \times 10^4 \text{ g mol}^{-1}$

58. Given below are two statements:

Statement-I: The radius of second orbit of He^+ is equal to that of first orbit of hydrogen.

Statement-II: The radius of an orbit is directly proportional to n^2 and inversely proportional to Z .

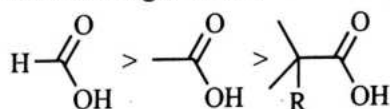
In the light of the above statements, choose the most appropriate answer from the options given below :

- (1) Statement-I is correct but Statement-II is incorrect.
- (2) Statement-I is incorrect but Statement-II is correct.
- (3) Both Statement-I and Statement-II are correct.
- (4) Both Statement-I and Statement-II are incorrect.

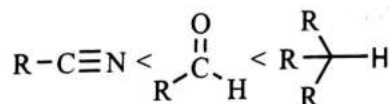
59. Consider the following statements;

(I) 2-Chloroethanoic acid is more acidic than ethanoic acid.

(II) Acidic strength order:



(III) Acidic strength order:



Select the **correct** statements;

- (1) I, II and III
- (2) I and II only
- (3) II and III only
- (4) I and III only

60. Which of the following pairs of compounds are isostructural?

- (1) XeF_2 and IF_2^-
- (2) NH_3 and BF_3
- (3) CO_3^{2-} and SO_3^{2-}
- (4) PCl_5 and ICl_5

61. Given below are two statement: one is labelled as Assertion (A) and the other is labelled as Reason (R)

Assertion (A): Addition of HBr to 1-butene gives two optical isomers.

Reason (R): The product contains one asymmetric carbon.

In the light of the above statements choose the **correct** answer from the options given below:

- (1) (A) is true but (R) is false.
- (2) (A) is false but (R) is true.
- (3) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (4) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

62. Which of the following nitrogen containing compounds give Lassaigne's test?

- A. Phenyl hydrazine
- B. Glycine
- C. Urea
- D. Hydrazine

- (1) A and B
- (2) B and C
- (3) Both A and D
- (4) A, B and C

63. Which of the following reactions will give an alkyne as a major product?

- (1) $\text{CH}_3\text{CBr}_2\text{CHBr}_2 \xrightarrow{\text{Zn dust}}$
- (2) $\text{CH}_3\text{CH}_2\text{CHBr}_2 \xrightarrow{\text{alc. KOH}} \text{A} \xrightarrow{\text{NaNH}_2} \text{B}$
- (3) $\text{CH}_3\text{CHBrCH}_2\text{Br} \xrightarrow[2 \text{ eq.}]{\text{NaNH}_2}$
- (4) All of these

64. When 5 mole of gas is heated at constant volume, temperature is raised from 410 K to 420 K. Heat supplied to the gas is 600 J. Then **correct** statement is;

- (1) $q = w = 600\text{J}$, $\Delta E = 0$
- (2) $q = \Delta E = 600\text{J}$, $w = 0$
- (3) $q = -w = 600\text{J}$, $\Delta E = 0$
- (4) $\Delta E = 0$, $q = w = -600\text{J}$

65. Which of the following transition metal have lowest first ionisation enthalpy?

- (1) Cr
- (2) Cu
- (3) Co
- (4) Fe

66. Which of the following compound is most stable among all?

- (1) 
- (2) 
- (3) 
- (4) 

67. In which of the following equilibrium, K_c and K_p are equal?

- (1) $2\text{C(s)} + \text{O}_2(\text{g}) \rightleftharpoons 2\text{CO}_2(\text{g})$
- (2) $\text{HI(g)} \rightleftharpoons \frac{1}{2}\text{H}_2(\text{g}) + \frac{1}{2}\text{I}_2(\text{g})$
- (3) $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightleftharpoons \text{CO}_2(\text{g}) + 2\text{H}_2\text{O(l)}$
- (4) $\text{XeF}_6(\text{g}) + \text{H}_2\text{O(g)} \rightleftharpoons \text{XeOF}_4(\text{g}) + 2\text{HF(g)}$

68. The **correct** order of the basic strength of methyl substituted amines in aqueous solution is;

- (1) $(\text{CH}_3)_2\text{NH} > \text{CH}_3\text{NH}_2 > (\text{CH}_3)_3\text{N}$
- (2) $(\text{CH}_3)_3\text{N} > \text{CH}_3\text{NH}_2 > (\text{CH}_3)_2\text{NH}$
- (3) $(\text{CH}_3)_3\text{N} > (\text{CH}_3)_2\text{NH} > \text{CH}_3\text{NH}_2$
- (4) $\text{CH}_3\text{NH}_2 > (\text{CH}_3)_2\text{NH} > (\text{CH}_3)_3\text{N}$

69. The decreasing order of boiling points of the following alcohols;

- (1) 3-Methylbutan-2-ol > 2-Methylbutan-2-ol > Pentan-1-ol
- (2) Pentan-1-ol > 3-Methylbutan-2-ol > 2-Methylbutan-2-ol
- (3) 2-Methylbutan-2-ol > 3-Methylbutan-2-ol > Pentan-1-ol
- (4) 2-Methylbutan-2-ol > Pentan-1-ol > 3-Methylbutan-2-ol

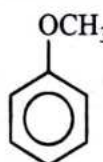
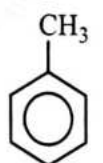
70. Select the **correct** order of boiling point;

- (1) $\text{PH}_3 < \text{AsH}_3 < \text{NH}_3 < \text{SbH}_3$
- (2) $\text{NH}_3 < \text{PH}_3 < \text{AsH}_3 < \text{SbH}_3$
- (3) $\text{PH}_3 < \text{AsH}_3 < \text{SbH}_3 < \text{NH}_3$
- (4) $\text{PH}_3 < \text{NH}_3 < \text{AsH}_3 < \text{SbH}_3$

71. The overall order of reaction between X and Y is 3. Which of the following rate equation must be **correct**, if on doubling the concentration of X, the rate of reaction gets doubled?

- (1) $r = K[\text{X}]^2[\text{Y}]^0$
- (2) $r = K[\text{X}]^1[\text{Y}]^2$
- (3) $r = K[\text{X}]^1[\text{Y}]^3$
- (4) $r = K[\text{X}]^2[\text{Y}]^1$

72. Which of the following compounds do **not** give Friedel craft reaction?

- (1) 
- (2) 
- (3) 
- (4) 

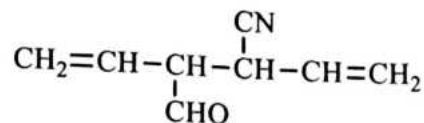
73. Which of the following is a fat soluble vitamin?

- (1) Vitamin B
- (2) Vitamin C
- (3) Vitamin B₁₂
- (4) Vitamin K

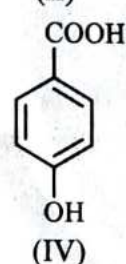
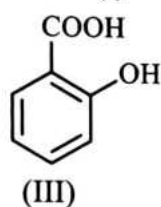
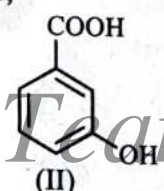
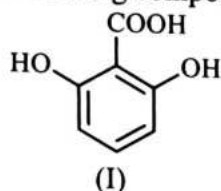
74. The number of significant figures in 0.00256 and 1.004 are respectively;

- (1) 3, 2
- (2) 5, 4
- (3) 3, 4
- (4) 6, 4

75. The **correct** IUPAC name of the following compound is;

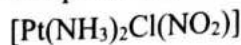


- (1) 3-cyano-2-ethenylpent-4-enal
(2) 2-ethenyl-3-formylpent-4-ene nitrile
(3) 2,3-diethenyl-4-oxobutane nitrile
(4) 3-cyano-4-formylhexa-1,5-diene
76. Invert sugar is an equimolar mixture of;
- (1) D-Glucose and D-Fructose
(2) D-Glucose and L-Fructose
(3) D-Glucose and L-Glucose
(4) D-Fructose and L-Fructose
77. The **correct** decreasing order of acidity for the following compounds is;



- (1) I > III > II > IV (2) III > I > II > IV
(3) III > IV > II > I (4) I > III > IV > I
78. At 80°C, the vapour pressure of pure liquid 'A' is 400 mm of Hg and that of pure liquid 'B' is 800 mm of Hg. If a solution of 'A' and 'B' boils at 80°C and 1 atm pressure, then amount of 'A' in the mixture is (1 atm = 760 mm of Hg);
- (1) 52% mole (2) 34% mole
(3) 10% mole (4) 48% mole
79. If the cell constant of the cell is 0.1 cm⁻¹ and resistance is 0.2, then conductance will be;
- (1) 0.02 Ω⁻¹ (2) 5 Ω⁻¹
(3) 0.05 Ω⁻¹ (4) 0.01 Ω⁻¹

80. The **correct** IUPAC name of the given compound is;



- (1) Diamminechloridonitrito-N-platinum(II)
(2) Diamminechloridonitrito-O-platinum(II)
(3) Diamminechloridonitrito-N-platinum(I)
(4) Diammoniachloridonitrito-O-platinum(II)
81. Cannizzaro reaction is **not** given by;
- (1) trimethylacetaldehyde
(2) formaldehyde
(3) benzaldehyde
(4) acetaldehyde
82. Oxalate ion is a;
- (1) anionic bidentate ligand.
(2) ambidentate ligand.
(3) neutral bidentate ligand.
(4) hexadentate ligand.

83. Match List-I with List-II.

List-I (Molecule)	List-II (Geometry)
A. PCl ₅	I. Trigonal planar
B. NH ₃	II. Octahedral
C. BCl ₃	III. Trigonal bipyramidal
D. SF ₆	IV. Tetrahedral

Choose the **correct** answer from the options given below:

- (1) A-III, B-IV, C-I, D-II
(2) A-III, B-I, C-II, D-IV
(3) A-III, B-IV, C-II, D-I
(4) A-II, B-III, C-I, D-IV
84. The half-life for the first order reaction is 69.3 sec. The time required for 80 g of the reactant to reduce to 20 g will be; (log 2 = 0.3)
- (1) 138.65 sec
(2) 138.65 × 10² sec
(3) 13.865 sec
(4) 13.865 × 10² sec

85. Given below are two statements:

Statement-I: N^{3-} has bigger size as compared to its neutral atom.

Statement-II: Al^{3+} has smaller size as compared to its neutral atom.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement-I is correct but Statement-II is incorrect.
- (2) Statement-I is incorrect but Statement-II is correct.
- (3) Both Statement-I and Statement-II are correct.
- (4) Both Statement-I and Statement-II are incorrect.

Chemistry : Section-B (Q. No. 86 to 100)

86. Given below are two statement: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Zn, Cd and Hg do not show the properties of transition elements.

Reason (R): Zn, Cd and Hg have completely filled valence shell electronic configuration.

In the light of the above **statements** choose the **correct** answer from the options given below:

- (1) (A) is true but (R) is false.
- (2) (A) is false but (R) is true.
- (3) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (4) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

87. $3A \rightarrow 2B$, rate of reaction $\frac{d[B]}{dt}$ is equal to;

- (1) $-\frac{3}{2} \frac{d[A]}{dt}$
- (2) $-\frac{2}{3} \frac{d[A]}{dt}$
- (3) $-\frac{1}{3} \frac{d[A]}{dt}$
- (4) $-2 \frac{d[A]}{dt}$

88. Given below are two statements :

Statement-I: The law of conservation of mass is valid for nuclear reaction.

Statement-II: As per law of conservation of mass in chemical reaction, the total mass of reactants and products remain same.

In the light of the above statements, choose the *most appropriate* answer from the options given below :

- (1) Statement-I is correct but Statement-II is incorrect.
- (2) Statement-I is incorrect but Statement-II is correct.
- (3) Both Statement-I and Statement-II are correct.
- (4) Both Statement-I and Statement-II are incorrect.

89. Which of the following is a colligative property?

- (1) Boiling point
- (2) Vapour pressure
- (3) Osmotic pressure
- (4) None of these

90. Oxidation state of central atom in CO_2 is;

- (1) +4
- (2) +2
- (3) +8
- (4) -2

91. Match the species given in Column-I with the chemical formula given in Column-II.

Column-I	Column-II
(a) Most abundant element	(i) B
(b) Lowest m. pt.	(ii) Tl
(c) Inert Pair Effect	(iii) Ga
(d) No d-orbitals	(iv) Al
(1) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)	
(2) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)	
(3) (a)-(iv), (b)-(i), (c)-(iii), (d)-(ii)	
(4) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)	

92. Identify the factor/s from the following that affects electrolytic conductance of a solution.

- A. The nature of the electrolyte added.
- B. The nature of the electrode used.
- C. Concentration of the electrolyte.
- D. The nature of solvent used.

- (1) Only (A) (2) Both (A) and (B)
(3) (B), (C) and (D) (4) (A), (C) and (D)

93. The correct order of stability of alkene is;

- (I) 1-hexene
- (II) cis-3-hexene
- (III) trans-3-hexene

- (1) II > III > I (2) III > II > I
(3) III > I > II (4) I > II > III

94. Given below are two statement: one is labelled as Assertion (A) and the other is labelled as Reason (R)

Assertion (A): First ionisation energy of nitrogen is lower than oxygen.

Reason (R): Across the period, effective nuclear-charge increases.

In the light of the above statements, choose the correct answer from the options given below:

- (1) (A) is true but (R) is false.
- (2) (A) is false but (R) is true.
- (3) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (4) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

95. In 6th period of the modern periodic table, electronic energy levels is arranged in the order;

- (1) $6s < 4f < 5d < 6p$
- (2) $6s < 6p < 4f < 5d$
- (3) $4f < 5d < 6s < 6p$
- (4) $6s < 5d < 4f < 6p$

96. Which of the molecule has maximum bond angle?

- (1) I_3^- (2) H_2O
- (3) BF_3 (4) NH_3

97. A basic buffer solution contains equal concentration of base and its salt. If the value of K_b is 10^{-5} , then the pH of the buffer solution would be;

- (1) 8 (2) 10
- (3) 12 (4) 9

98. Match List-I with List-II.

List-I (Electrolyte)	List-II (Van't Hoff factor)
A. NaCl	I. 5
B. $C_6H_{12}O_6$	II. 2
C. K_2SO_4	III. 1
D. $Al_2(SO_4)_3$	IV. 3

Choose the correct answer from the options given below:

- (1) A-II, B-III, C-IV, D-I
- (2) A-III, B-II, C-IV, D-I
- (3) A-II, B-III, C-I, D-IV
- (4) A-I, B-III, C-IV, D-II

99. In which of the following property, trans isomer is greater than cis isomer?

- (1) Boiling point (2) Solubility
- (3) Melting point (4) Dipole moment

100. Which one is least reactive towards S_N1 reaction?

- (1) CH_3-Cl
- (2) $(CH_3)_3C-Cl$
- (3) $CH_2=CH-CH_2-Cl$
- (4) $C_6H_5CH_2Cl$

Botany : Section-A (Q. No. 101 to 135)

101. Which of the following statements are not correct?

- I. Lower the taxon, more are the characteristics that the members within the taxon share.
- II. Order is the assemblage of genera which exhibit a few similar characters.
- III. Cat and dog are included in the same family Felidae.
- IV. Binomial nomenclature was introduced by Carolus Linnaeus.

- (1) I, II and III (2) II, III and IV
- (3) I and IV (4) II and III

102. How many of the given fungi produce conidia as an asexual spore?

Bracket fungi, *Claviceps*, Morels, *Ustilago*,
Alternaria, *Agaricus*, Puffballs,
Colletotrichum, *Aspergillus*

- (1) Three (2) Four
 (3) Six (4) Two

103. Given below are two statements:

Statement-I: A flower is a modified shoot wherein the shoot apical meristem changes to floral meristem.

Statement-II: When a shoot tip transforms into a flower, it is never solitary.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement-I is correct but Statement-II is incorrect.
 (2) Statement-I is incorrect but Statement-II is correct.
 (3) Both Statement-I and Statement-II are correct.
 (4) Both Statement-I and Statement-II are incorrect.

104. Find out the incorrect option about heterochromatin.

- (1) It is densely packed.
 (2) It stains lightly.
 (3) It is transcriptionally inactive.
 (4) Present in nucleus.

105. Adventitious roots are found in all the following, except:

- (1) grass. (2) *Monstera*.
 (3) banyan tree. (4) carrot.

106. Isogamous condition with flagellated (motile) gametes is found in algae:

- (1) *Spirogyra*. (2) *Volvox*.
 (3) *Fucus*. (4) *Ulothrix*.

107. Mark the incorrect.

- (1) Truffles are edible fungi
 (2) *Rhizopus* is bread mould
 (3) *Albugo* is parasitic on plants
 (4) Conidia is present in *Ustilago*

108. Match the type of fruit with their respective plant family:

Type of Fruits		Plant family	
A.	The fruit is often a capsule that splits open (schizocarp) or a nut	(i)	Cruciferae
B.	The fruit is a silique or silicle, which is a long, narrow pod that splits open to release seeds.	(ii)	Leguminosae
C.	The fruit is a legume (pod) that splits open along both sutures to release seeds.	(iii)	Gramineae
D.	The fruit can be an achene, a cypsela, or a pappus, depending on the individual florets.	(iv)	Malvaceae
E.	The fruit is typically a grain or caryopsis, which is a one-seeded fruit with the seed fused to the ovary wall.	(v)	Compositae

- (1) A-(v), B-(iv), C-(iii), D-(ii), E-(i)
 (2) A-(i), B-(ii), C-(iii), D-(iv), E-(v)
 (3) A-(iv), B-(i), C-(ii), D-(v), E-(iii)
 (4) A-(iii), B-(i), C-(ii), D-(v), E-(iv)

109. Arrange the following components of an ecosystem in the order of energy flow:

- Producers
- Secondary consumers
- Primary consumers
- Decomposers
- Sun

Options:

- (e)-(a)-(c)-(b)-(d)
- (e)-(a)-(b)-(c)-(d)
- (a)-(b)-(c)-(d)-(e)
- (d)-(c)-(b)-(a)-(e)

110. Match List-I with List-II:

List-I	List-II
A. Dicotyledonous Root	I. Sclerenchymatous hypodermis
B. Monocotyledonous Root	II. Collenchymatous hypodermis
C. Dicotyledonous Stem	III. More than six xylem bundle & pith is well developed
D. Monocotyledonous Stem	IV. The pith is small or inconspicuous

Choose the **correct** answer from the options given below:

- A-III; B-II; C-IV; D-I
- A-III; B-II; C-I; D-IV
- A-IV; B-III; C-II; D-I
- A-II; B-I; C-III; D-IV

111. Given below are two statements:

Statement-I: Robert Brown first saw and described a live cell.

Statement-II: Anton Von Leeuwenhoek discovered the nucleus.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

- Statement-I is correct but Statement-II is incorrect.
- Statement-I is incorrect but Statement-II is correct.
- Both Statement-I and Statement-II are correct.
- Both Statement-I and Statement-II are incorrect.

112. Correct statements are

- The excreta of cattle, commonly called gobar, is rich in *Rhizobium*.
- In rumen, methanogens help in breakdown of cellulose.
- Methanogens play an important role in the nutrition of cattle.
- Methane, CO₂ and traces of nitrogen, hydrogen sulphide, and hydrogen make up biogas.

- (a) & (b)
- (b), (c) & (d)
- (a) & (d)
- (b) & (c)

113. Choose the **incorrectly** matched pair.

- Tracheids- one of the main water transporting elements.
- Vessel- characteristic feature of angiosperms.
- Xylem fibres- highly thickened walls.
- Xylem parenchyma- provide mechanical support.

114. Match List-I with List-II:

List-I	List-II
A. Amyloplasts	I. Store proteins
B. Chromoplasts	II. Store oil and fats
C. Aleuroplasts	III. Store carbohydrates
D. Elaioplasts	IV. Carotenoid pigments

Choose the **correct** answer from the options given below:

- A-II; B-III; C-IV; D-I
- A-III; B-IV; C-I; D-II
- A-IV; B-III; C-II; D-I
- A-III; B-II; C-I; D-IV

115. If you have obtained both tall and dwarf pea plants in the same proportions in a cross, which of the following combinations of pea plant genotypes would have been used?

- TT × tt
- Tt × Tt
- TT × Tt
- tt × tt

116. Given below are two statements:

Statement-I: In anaerobic organisms, glycolysis is the only process in respiration.

Statement-II: Glycolysis occurs in all living organisms.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement-I is correct but Statement-II is incorrect.
- (2) Statement-I is incorrect but Statement-II is correct.
- (3) Both Statement-I and Statement-II are correct.
- (4) Both Statement-I and Statement-II are incorrect.

117. Match List-I with List-II:

List-I		List-II
A. Leptotene	I.	Dissolution of the synaptonemal complex
B. Zygotene	II.	Appearance of recombination nodules
C. Pachytene	III.	Formation of complex structure called synaptonemal complex
D. Diplotene	IV.	Compaction of chromosomes continues throughout this stage

Choose the **correct** answer from the options given below:

- (1) A-IV; B-III; C-II; D-I
- (2) A-III; B-IV; C-I; D-II
- (3) A-I; B-II; C-III; D-IV
- (4) A-II; B-III; C-I; D-IV

118. Which radioactive isotopes were used by Hershey and Chase (1952) in their experiments?

- (1) ^{35}S and ^{15}N
- (2) ^{32}P and ^{35}S
- (3) ^{32}P and ^{15}N
- (4) ^{14}N and ^{15}N

119. Select the **incorrect** statement(s) from the following.

- A. The 'bakanae' (foolish seedling) disease of rice seedlings, was caused by a fungal pathogen *Gibberella fujikuroi*. The active substances were later identified as auxin.
 - B. Auxins control xylem differentiation and help in cell division.
 - C. Miller et al. (1955) identified and crystallised the cytokinesis promoting active substance that they termed kinetin.
 - D. Cousins confirmed the release of a volatile substance from ripened oranges that hastened the ripening of stored unripened bananas. Later it was identified as ethylene, a gaseous PGR.
- (1) A only (2) B and C
(3) C and D (4) D only

120. The electrons that reduce NADP^+ to $\text{NADPH} + \text{H}^+$ during light reaction are directly from:

- (1) photosystem I.
- (2) water.
- (3) carbon dioxide.
- (4) photosystem II.

121. The cell organelle which is double membrane bound, site of aerobic respiration, and produces cellular energy in the form of ATP is:

- (1) chloroplast.
- (2) ribosomes.
- (3) mitochondria.
- (4) E.R.

122. Choose the **incorrect** statement (s) w.r.t. the causes of biodiversity losses.

- A. Many species like Steller's Sea cow, passenger pigeon became extinct in the last 500 years due to overexploitation by humans.
- B. Nile perch & species of cichlid fish are examples of co-extinctions.
- C. In a coevolved plant-pollinator mutualism, extinction of one invariably leads to the extinction of the other species.
- D. Introduction of the *Clarias gariepinus* for aquaculture purposes is posing a threat to the indigenous catfishes in our rivers.

- (1) A and B (2) B only
- (3) B and C (4) C and D

123. Which of the following is **not** a characteristic feature during mitosis in somatic cells?

- (1) Spindle fibres
- (2) Disappearance of nucleolus
- (3) Chromosomes movement
- (4) Synapsis

124. Oxidative phosphorylation is formation of:

- (1) NADPH + H^+ in photosynthesis.
- (2) ATP in photosynthesis.
- (3) NADPH + H^+ in respiration.
- (4) ATP in respiration.

125. _____ was isolated by F.W. Went from tips of coleoptiles of oat seedlings.

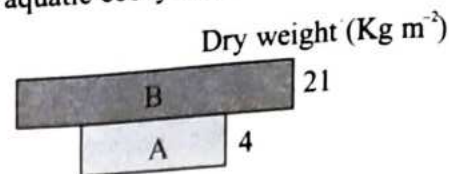
- (1) Absciscic acid
- (2) Ethylene
- (3) Gibberellins
- (4) Auxin

126. Read the following and identify the **incorrect** statement(s).

- A. Meiosis involves two sequential cycles of nuclear and cell division called meiosis I and meiosis II but only a single cycle of DNA replication.
- B. Meiosis I is initiated after the parental chromosomes have replicated to produce identical sister chromatids at the S phase.
- C. Meiosis involves pairing of homologous chromosomes and recombination between sister chromatids of homologous chromosomes.
- D. Four haploid cells are formed at the end of meiosis II.

- (1) A and B
- (2) C and D
- (3) C only
- (4) B and D

127. Given figure represents a pyramid of biomass in an aquatic ecosystem.



Identify A and B and select the correct answer.

- (a) A is crop which supports and B is the crop which is supported.
 - (b) A is the crop which is supported and B is the crop which supports.
 - (c) A is phytoplanktons and B is zooplanktons.
 - (d) A is zooplanktons and B is phytoplanktons.
- (1) (a) and (d)
 - (2) (b) and (c)
 - (3) (a) and (c)
 - (4) (b) and (d)

128. Kernel colour in corn is a trait determined by two alleles. The dominant allele (P) produces a purple colour and the recessive allele (p) produces a yellow colour. By crossing two plants, an ear of corn is produced. The kernels are purple and yellow coloured.

The yellow kernels can be best described as

- (1) Homozygous dominant
- (2) Hybrid
- (3) Heterozygous
- (4) Homozygous recessive

129. Select the **incorrect** statement w.r.t. Calvin cycle.

- (1) Carboxylation is the most crucial step of the Calvin cycle where CO_2 is utilised for the carboxylation of RuBP.
- (2) Carboxylation is catalysed by the enzyme RuBP carboxylase.
- (3) Reduction in Calvin cycle is a series of reactions that lead to the formation of 3-PGA.
- (4) The regeneration steps require one ATP for phosphorylation to form RuBP.

130. Consider the detailed steps in the process of megasporogenesis in flowering plants leading to the formation of the embryo sac:

- MMC undergoes meiosis to produce four megaspores.
- Three megaspores degenerate, leaving one functional megaspore.
- Functional megaspore's nucleus divides mitotically, creating a 2-nucleate embryo sac.
- Nuclei undergo successive mitotic divisions to reach 4-nucleate and then 8-nucleate stages.
- Out of 8 nuclei, 6 form cell walls, creating distinct cells; two polar nuclei remain free in the central cell.

Choose the correct sequence of these events from the options given below to accurately depict the monosporic development pathway:

- (a)-(b)-(c)-(d)-(e)
- (b)-(a)-(c)-(d)-(e)
- (c)-(a)-(b)-(e)-(d)
- (d)-(c)-(b)-(a)-(e)

131. Which of the following statements about sporopollenin is **incorrect**?

- Exine is made up of sporopollenin.
- Sporopollenin is one of the most resistant organic materials.
- Exine has apertures called germ pores where sporopollenin is present.
- Sporopollenin can withstand high temperatures and strong acids.

132. Read the following statements carefully. Identify and mark the pre-fertilisation events illustrated.

- A haploid parent produces gamete by mitosis.
- Zygote undergoes cell division during embryogenesis.
- In seed plants, pollen grains are the carriers of male gamete and ovule have the egg.
- Most vital event of sexual reproduction is fusion of gametes.
- Ovary wall proliferates to form pericarp.

- (a), (c) & (d)
- (b) & (d)
- (a), (b) & (e)
- (c), (d) & (e)

133. In the process of fertilization, the pollen tube of a pollen grain typically gains access to the embryo sac through which specific region to ensure successful fertilization?

- Integument
- Nucellus
- Chalaza
- Micropyle

134. Match the following floral formula with their respective floral diagram and example:

Floral family	Floral Diagram	Example
(a) Malvaceae	1.	(i) <i>Hibiscus rosasinensis</i>
(b) Brassicaceae	2.	(ii) <i>Brassica rapa</i>
(c) Abaceae	3.	(iii) <i>Lathyrus odoratus</i>
(d) Solanaceae	4.	(iv) <i>Solanum nigrum</i>
(e) Gramineae	5.	(v) <i>Triticum aestivum</i>
(f) Compositae	6.	(vi) <i>Helianthus annuus</i>

- (1) (a)-2-(i), (b)-1-(ii), (c)-4-(iii), (d)-3-(iv),
(e)-6-(v), (f)-5-(vi)
- (2) (a)-3-(ii), (b)-2-(i), (c)-1-(vi), (d)-4-vi,
(e)-(5)-(v), (f)-6-(iii)
- (3) (a)-1-(iii), (b)-6-(ii), (c)-5-(i), (d)-2-(vi),
(e)-3-(v), (f)-4-(iv)
- (4) (a)-5-(iv), (b)-4-(ii), (c)-3-(iii), (d)-6-(iv),
(e)-1-(vi), (f)-(2)-(v)

135. Match List-I containing characters with List-II containing recessive trait studied by Mendel in Pea plants

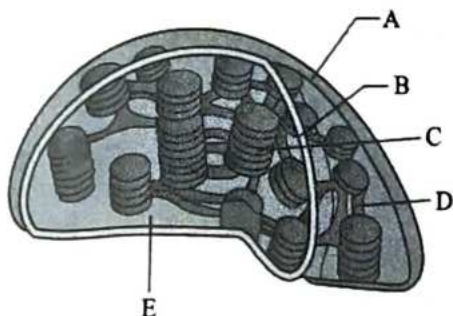
List-I (Characters)	List-II (Recessive trait)
A. Pod colour	I. Yellow
B. Seed colour	II. Green
C. Seed shape	III. Wrinkled
D. Pod shape	IV. Constricted

Choose the **correct** answer from the options given below:

- (1) A-IV; B-III; C-II; D-I
(2) A-III; B-IV; C-II; D-I
(3) A-I; B-II; C-III; D-IV
(4) A-II; B-I; C-III; D-IV

Botany : Section-B (Q. No. 136 to 150)

136. Consider the following five statements (A to E) w.r.t. chloroplast shown below. Select the correct option stating which ones are True (T) and which ones are False (F)



- A. It is impermeable
B. It is relatively less permeable
C. It is the site of production of assimilatory power
D. Present between two grana and contains enzymes of dark reaction
E. It contains enzymes for the synthesis of sugar and proteins

	A	B	C	D	E
(1)	F	T	T	T	T
(2)	F	T	T	F	T
(3)	T	F	T	T	T
(4)	T	F	F	T	T

137. Match List-I with List-II:

List-I	List-II
A. MacLeod and McCarty	I. <i>Lac</i> operon
B. Hershey and Chase	II. Semi-conservative DNA replication
C. Meselson and Stahl	III. Proved DNA as genetic material
D. Jacob and Monod	IV. Biochemical nature of transforming principle

Choose the **correct** answer from the options given below:

- (1) A-IV; B-III; C-II; D-I
(2) A-III; B-IV; C-II; D-I
(3) A-IV; B-II; C-III; D-I
(4) A-II; B-I; C-III; D-IV

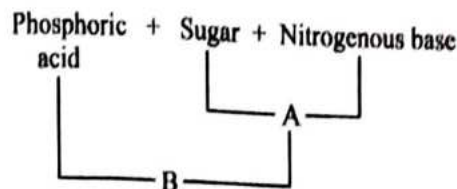
138. In a Mendelian cross, the F_2 generation had both genotypic and phenotypic ratios of 1 : 2 : 1. What **type** of genetic inheritance pattern does it represent?

- (1) Co-dominance
(2) Dihybrid cross
(3) Monohybrid cross with complete dominance
(4) Monohybrid cross with incomplete dominance

139. _____ is present in grasses and regenerate parts removed by the grazing herbivores.

- (1) Root apical meristem
(2) Intercalary meristem
(3) Shoot apical meristem
(4) Lateral meristem

140. What does A and B represent in the given representation?



- (1) A - Nucleotide, B - Nucleoside
 (2) A - Nucleoside, B - Nucleotide
 (3) A - Nucleoside, B - Deoxyribonucleoside
 (4) A - Nucleotide, B - Deoxyribonucleotide
141. Given below are two statement: one is labelled as Assertion (A) and the other is labelled as Reason (R)
Assertion (A): Most of the photosynthesis takes place in the blue and red regions of the spectrum.
Reason (R): Chlorophyll *a* is the chief pigment associated with photosynthesis.
 In the light of the above statements, choose the correct answer from the options given below:
- (1) (A) is true but (R) is false.
 (2) (A) is false but (R) is true.
 (3) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (4) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

142. Match List-I with List-II w.r.t. Lac operon:

List-I	List-II
A. <i>a</i> gene	I. Beta-galactosidase
B. <i>z</i> gene	II. Transacetylase
C. Regulator gene	III. Codes for protein/enzyme
D. Structural gene	IV. Codes for repressor molecule

Choose the correct answer from the options given below:

- (1) A-II; B-I; C-III; D-IV
 (2) A-II; B-I; C-IV; D-III
 (3) A-IV; B-III; C-I; D-II
 (4) A-II; B-III; C-I; D-IV

143. Match the List-I and List-II:

List-I	List-II
A. Nitrogen-fixing cyanobacteria	I. <i>Azospirillum</i>
B. Symbiotic nitrogen fixer	II. <i>Rhizobium</i>
C. Free-living nitrogen fixer	III. <i>Anabaena</i>

Choose the correct answer from the options given below:

- (1) A-I; B-II; C-III
 (2) A-I; B-III; C-II
 (3) A-II; B-III; C-I
 (4) A-III; B-II; C-I

144. Select the correct pair w.r.t. given terms:

- (1) Salinity – Sugar concentration in water
 (2) Suspensions – Change in morphological structure
 (3) Phytophagous – Insect feeding on plant sap and plant part
 (4) Sexual deceit – Fig flowers

145. Given below are two statement: one is labelled as Assertion (A) and the other is labelled as Reason (R)

Assertion (A): Ethylene promotes abscission of plant organs especially of leaves and flowers.

Reason (R): Ethylene inhibits the rapid internode/petiole elongation in deep water rice plants.

In the light of the above statements, choose the correct answer from the options given below:

- (1) (A) is true but (R) is false.
 (2) (A) is false but (R) is true.
 (3) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (4) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

146. Select the **correct** statement(s) from the following.
- IUCN stands for International Union for Conservation of Nature and Narcotics Resources.
 - According to IUCN (2004), the total number of plant and animal species described so far is less than 1.5 million.
 - Less than 22 per cent of all the species recorded are animals, while plants comprise no more than 70 per cent of the total.
 - Robert May places the global species diversity at about 7 million.
- (1) A only (2) B and C
(3) C and D (4) D only
147. Select the mis-matched pair w.r.t. given plants.
- Night blooming flower – Usually white in colour
 - Cactus – Large leaves
 - Calotropis* – Cardiac glycoside
 - Acacia* – Sharp thorns
148. Identify the **correct** statement(s) from the following.
- Carrot-grass and water hyacinth became extinct due to co-extinction.
 - Passenger pigeon became extinct due to alien species invasion.
 - More than 200 species of cichlid fish became extinct due to the introduction of the Nile perch into Lake Victoria in east Africa.
 - African catfish *Clarias gariepinus* became extinct due to the indigenous catfishes in our rivers.
- (1) A only (2) B and C
(3) C only (4) B and D
149. In 1953, James Watson and Francis Crick proposed the double helix model of DNA, a pivotal moment in the history of biology. Which of the following contributed to their model?

- X-ray diffraction studies of DNA done by Wilkins and Franklin.
- Chargaff's rule.
- Griffith's transformation experiment.
- Meselson and Stahl's experiment.

Choose the *most appropriate* answer from the options given below:

- I, II and IV only
- I and II only
- I, II, III and IV
- III and IV only

150. The 'Evil Quartet' includes all the following, **except**:

- habitat loss and fragmentation.
- over-exploitation.
- species diversity.
- co-extinctions.

Zoology : Section-A (Q. No. 151 to 185)

151. Which of the following options have all sexually transmitted diseases?

- AIDS, Syphilis, Cholera
- Trichomoniasis, Malaria, AIDS
- Gonorrhoea, Hepatitis B, Chlamydiasis
- Hepatitis B, AIDS, Haemophilia

152. Match List-I with List-II:

List-I	List-II
A. Human chorionic gonadotropin	I. Pituitary
B. Relaxin	II. Placenta
C. Oxytocin	III. Leydig cells
D. Androgens	IV. Ovary

Choose the **correct** answer from the options given below:

- A-IV, B-I, C-II, D-III
- A-I, B-II, C-IV, D-III
- A-III, B-II, C-IV, D-I
- A-II, B-IV, C-I, D-III

153. Which of the following shows convergent evolution?

- (1) Lemur and Marsupial mouse
- (2) Bobcat and Tasmanian tiger cat
- (3) Numbat and Marsupial mole
- (4) Anteater and Tasmanian wolf

154. The most abundant enzyme in the animal world is:

- (1) collagen.
- (2) RuBisCO.
- (3) trypsin.
- (4) insulin.

155. Sequence the following steps in an IVF cycle for embryo transfer:

- (a) Assessment of embryo quality
- (b) Fertilization and embryo culture in the lab
- (c) Hormonal stimulation of the ovaries
- (d) Transfer of selected embryo into the uterus
- (e) Retrieval of multiple oocytes

Choose the correct sequence from the options given below:

- (1) (d)-(a)-(e)-(b)-(c)
- (2) (b)-(a)-(d)-(e)-(c)
- (3) (c)-(a)-(d)-(e)-(b)
- (4) (c)-(e)-(b)-(a)-(d)

156. Which of the following statements are correct?

- A. Pteridophytes form coal deposits.
- B. The first organisms that invaded land were shrews.
- C. Lichens can be used as industrial pollution indicators.
- D. *Tyrannosaurus rex* was the biggest dinosaur, which was about 20 feet in height.

Choose the *most appropriate* answer from the options given below:

- (1) A, B and C only
- (2) B, C and D only
- (3) A, C and D only
- (4) A, B and D only

157. The midbrain of the frog is characterised by a pair of:

- (1) cerebral hemisphere.
- (2) cerebellum.
- (3) optic lobes.
- (4) olfactory lobes.

158. The following statements show some characteristics of an organism. Identify it.

- A. It has an aquatic form.
- B. It also possesses lateral appendages.
- C. It is dioecious.

- (1) *Ancylostoma* (Hookworm)
- (2) *Nereis*
- (3) *Chaetopleura*
- (4) *Clarias* (Magur)

159. Given below are two statements:

Statement-I: Oogenesis is initiated during the embryonic development stage.

Statement-II: Zona pellucida is a new membrane formed by the primary oocyte.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement-I is correct but Statement-II is incorrect.
- (2) Statement-I is incorrect but Statement-II is correct.
- (3) Both Statement-I and Statement-II are correct.
- (4) Both Statement-I and Statement-II are incorrect.

160. Match List-I with List-II:

List-I	List-II
A. <i>Epidermophyton</i>	I. Ringworms
B. <i>Wuchereria</i>	II. Pneumonia
C. <i>Salmonella</i>	III. Elephantiasis
D. <i>Streptococcus</i>	IV. Typhoid

Choose the **correct** answer from the options given below:

- (1) A-I, B-III, C-IV, D-II
- (2) A-I, B-II, C-IV, D-III
- (3) A-III, B-II, C-IV, D-I
- (4) A-II, B-IV, C-I, D-III

161. The process of release of urine, i.e., micturition, causes due to:

- (1) relaxation of smooth muscles of the bladder and contraction of the urethral sphincter.
- (2) contraction of smooth muscles of the bladder and relaxation of the urethral sphincter.
- (3) contraction of skeletal muscles of the bladder and relaxation of the urethral sphincter.
- (4) relaxation of skeletal muscles of the bladder and contraction of the urethral sphincter.

162. Given below are two statements:

Statement-I: Starch cannot hold I_2 molecules in the helical portion.

Statement-II: Cellulose does not contain complex helices and hence can hold I_2 .

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement-I is correct but Statement-II is incorrect.
- (2) Statement-I is incorrect but Statement-II is correct.
- (3) Both Statement-I and Statement-II are correct.
- (4) Both Statement-I and Statement-II are incorrect.

163. Match List-I with List-II:

List-I	List-II (pO_2 of mm Hg)
A. Atmospheric air	I. 40
B. Oxygenated blood	II. 104
C. Alveoli	III. 159
D. Tissue	IV. 95

Choose the **correct** answer from the options given below:

- (1) A-III, B-IV, C-II, D-I
- (2) A-II, B-I, C-IV, D-III
- (3) A-I, B-IV, C-II, D-III
- (4) A-III, B-I, C-II, D-IV

164. Read the following statements:

- A. In vaccination, a preparation of antigenic proteins of pathogen or inactivated/weakened pathogen (vaccine) are introduced into the body.
- B. The vaccines can generate memory cells.
- C. Antitoxins are preparations containing antibodies to the toxin.
- D. Hepatitis B vaccine is produced from viruses.

Choose the option with the **correct** statement(s):

- (1) A only
- (2) A, B and C
- (3) C and D
- (4) D only

165. Select the **incorrect** statement w.r.t. the circulatory system in humans.

- (1) The sino-atrial node can generate the maximum number of action potentials.
- (2) Blood pressure in the pulmonary artery is more than that in the pulmonary vein.
- (3) The hepatic portal vein drains blood to the liver from the intestine.
- (4) Ventricular systole increases the flow of blood into the ventricles from the atria by about 30 percent.

166. Match List-I with List-II:

List-I	List-II
A. Retroviruses	I. pBR322
B. <i>Hind III</i>	II. Join DNA fragments
C. Kanamycin	III. Vectors
D. Ligase	IV. Antibiotics



Choose the **correct** answer from the options given below:

- (1) A-III, B-I, C-IV, D-II
- (2) A-II, B-III, C-I, D-IV
- (3) A-I, B-III, C-II, D-IV
- (4) A-III, B-IV, C-II, D-I

167. Given below are two statements: one is labeled as Assertion A and the other is labeled as Reason R:
Assertion (A): Members of the phylum Platyhelminthes are also called flatworms.

Reason (R): They have dorso-ventrally flattened bodies.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) (A) is true but (R) is false.
- (2) (A) is false but (R) is true.
- (3) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (4) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

168. Choose the **incorrect** statement w.r.t. secondary metabolites.

- (1) Abrin and ricin are lectins.
- (2) Rubber and drugs are useful for human welfare.
- (3) Codeine is an alkaloid.
- (4) Anthocyanins are pigments.

169. Which of the following option belongs exclusively to Cnidarians?

- (1) *Aplysia*, *Gorgonia*, *Spongilla* and *Hydra*
- (2) *Hydra*, *Pennatula*, *Obelia* and *Echinus*
- (3) *Hydra*, *Obelia*, *Adamsia* and *Physalia*
- (4) *Scypha*, *Hydra*, *Adamsia* and *Physalia*

170. Match List-I with List-II:

List-I	List-II
A. Tight junctions	I. Cementing to keep neighbouring cells together
B. Adhering junctions	II. Stop substances from leaking across a tissue
C. Gap junctions	III. Neurotransmitters
D. Synapses	IV. Cell to cell communication by connecting their cytoplasm

Choose the **correct** answer from the options given below:

- (1) A-I, B-III, C-II, D-IV
- (2) A-II, B-I, C-IV, D-III
- (3) A-IV, B-II, C-I, D-III
- (4) A-I, B-III, C-IV, D-II

171. C-peptide of human insulin is:

- (1) a part of the mature insulin molecule.
- (2) responsible for the formation of disulphide bridges.
- (3) removed during the maturation of pro-insulin to insulin.
- (4) responsible for its biological activity.

172. Given below are two statements: one is labeled as Assertion A and the other is labeled as Reason R:
Assertion (A): *E. coli* having pBR322 with DNA insert at the *BamH I* site can grow in medium containing tetracycline.

Reason (R): Recognition site for the *BamH I* is present in the tetracycline region of pBR322.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) (A) is true but (R) is false.
- (2) (A) is false but (R) is true.
- (3) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (4) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

173. Graves' disease is caused due to:

- (1) hyposecretion of thyroid gland.
- (2) hypersecretion of thyroid gland.
- (3) hyposecretion of adrenal gland.
- (4) hypersecretion of adrenal gland.

174. Match List-I with List-II:

List-I	List-II
A. Tunica media	I. End of systole
B. Tunica externa	II. Smooth muscle
C. Number of QRS complexes	III. Collagen fibre
D. T-wave	IV. Heart beat rate

Choose the **correct** answer from the options given below:

- (1) A-IV, B-I, C-II, D-III
- (2) A-II, B-I, C-IV, D-III
- (3) A-II, B-III, C-IV, D-I
- (4) A-IV, B-II, C-III, D-I

175. Suppose an electrical current has moved DNA fragments through a gel. What does the band on this gel farthest from the top (from where the DNA fragments were added) represent?

- (1) Shortest fragments of DNA.
- (2) Longest fragments of DNA.
- (3) All the fragments are found near the well only.
- (4) All of these

176. The urethra originates from the urinary bladder and extends through the penis to its external opening is called:

- (1) seminal vesicle.
- (2) hymen.
- (3) urethral meatus.
- (4) glans penis.

177. Match List-I with List-II:

List-I	List-II
A. Invertebrates	I. 500 mya
B. Jawless fish	II. 320 mya
C. Sea weeds	III. 350 mya
D. <i>Dryopithecus</i>	IV. 15 mya

Choose the **correct** answer from the options given below:

- (1) A-I, B-III, C-II, D-IV
- (2) A-II, B-III, C-IV, D-I
- (3) A-IV, B-II, C-I, D-III
- (4) A-II, B-IV, C-I, D-III

178. Given below are two statements:

Statement-I: RNAi is a method of cellular defense.

Statement-II: *Bacillus thuringiensis* produces protein that kills lepidopteran insects only.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

- (1) Statement-I is correct but Statement-II is incorrect.
- (2) Statement-I is incorrect but Statement-II is correct.
- (3) Both Statement-I and Statement-II are correct.
- (4) Both Statement-I and Statement-II are incorrect.

179. Spirometer can assess all of these lung volumes, **except**:

- (1) inspiratory reserve volume.
- (2) expiratory reserve volume.
- (3) residual volume.
- (4) tidal volume.

180. Corpora quadrigemina are present on _____ portion of the midbrain.

- (1) dorsal
- (2) ventral
- (3) lateral
- (4) posterior

181. Match List-I with List-II:

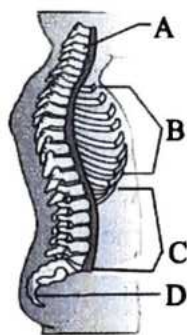
List-I	List-II
A. Gametogenesis	I. Transfer of sperms into female genital tract
B. Insemination	II. Formation of gametes
C. Gestation	III. Delivery of baby
D. Parturition	IV. Embryonic development

Choose the **correct** answer from the options given below:

- (1) A-IV, B-I, C-II, D-III
- (2) A-II, B-I, C-IV, D-III
- (3) A-II, B-III, C-IV, D-I
- (4) A-IV, B-II, C-III, D-I

182. Which of the following represents a palindromic sequence in DNA?

- (1) 5'-GAATTC-3'
3'-CTTAAG-5'
- (2) 5'-CCAATG-3'
3'-CAATTC-5'
- (3) 5'-CATTAG-3'
3'-GATAAC-5'
- (4) 5'-GATACC-3'
3'-CCTAAG-5'

183. Select the **correct** option regarding the following diagram.

- (1) A-Cervical vertebra-7
- (2) B-Lumbar vertebra-12
- (3) C-Thoracic vertebra-5
- (4) D-Coccyx vertebra- 6

184. Match List-I with List-II:

List-I	List-II
A. Malleus, Incus, Stapes	I. Bones of forelimb
B. Hyoid	II. Ear ossicles
C. Humerus, Radius, Ulna	III. U-shaped bone
D. Patella	IV. Knee cap

Choose the **correct** answer from the options given below:

- (1) A-I, B-III, C-II, D-IV
- (2) A-II, B-III, C-I, D-IV
- (3) A-II, B-IV, C-III, D-I
- (4) A-IV, B-II, C-III, D-I

185. Select the **correct** statement.

- A. PCR is used to detect HIV in suspected AIDS patients.
- B. Early detection by PCR is not possible.
- C. Pomato is a somatic hybrid of tomato and potato.
- D. A probe is not tagged with a radioactive molecule.

Choose the **most appropriate** answer from the options given below:

- (1) A and C only
- (2) C and D only
- (3) B and C only
- (4) A and D only

Zoology : Section-B (Q. No. 186 to 200)

186. Which of the following contraceptive methods is most suitable for couples seeking long-term, reversible contraception with maximum reliability if properly used?

- (a) Lactational amenorrhea
- (b) Tubectomy
- (c) Intrauterine devices (IUDs)
- (d) Implants
- (e) Emergency contraceptives

Choose the **correct** option:

- (1) (a) and (b)
- (2) (b) and (d)
- (3) (c) and (d)
- (4) (a) and (e)

187. Match List-I with List-II:

List-I	List-II
A. Thyroxine	I. Iodothyronine
B. Epinephrine	II. Amino-acid derivatives
C. Estradiol	III. Peptide hormone
D. Glucagon	IV. Steroid hormones

Choose the **correct** answer from the options given below:

- (1) A-I, B-II, C-IV, D-III
- (2) A-II, B-III, C-IV, D-I
- (3) A-I, B-II, C-III, D-IV
- (4) A-I, B-IV, C-II, D-III

188. Which of the following are **incorrect**?

- A. Rate of a physical or chemical process refers to the amount of product formed per unit time.
- B. Rate of reaction can also be called velocity if the direction is not specified.
- C. Inorganic catalysts work efficiently at high temperatures.
- D. All enzymes are proteins.

Choose the **most appropriate** answer from the options given below:

- (1) A and C only
- (2) B and C only
- (3) B and D only
- (4) A and D only

189. Match List-I with List-II:

List-I	List-II
A. Non-medicated intrauterine device	I. Lippes loop
B. Oral pill	II. Nirodh
C. Male condom	III. LNG-20
D. Hormone releasing intrauterine device	IV. Saheli

Choose the **correct** answer from the options given below:

- (1) A-III, B-I, C-IV, D-II
- (2) A-II, B-I, C-IV, D-III
- (3) A-I, B-IV, C-II, D-III
- (4) A-III, B-II, C-IV, D-I

190. Given below are two statements:

Statement-I: *Homo erectus* had a large brain around 1400cc.

Statement-II: Disturbance in Hardy-Weinberg equilibrium results in evolution.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

- (1) Statement-I is correct but Statement-II is incorrect.
- (2) Statement-I is incorrect but Statement-II is correct.
- (3) Both Statement-I and Statement-II are correct.
- (4) Both Statement-I and Statement-II are incorrect.

191. Match List-I with List-II:

List-I	List-II
A. <i>cryIac</i>	I. Emphysema
B. <i>cryIab</i>	II. Cotton bollworm
C. alpha-lactalbumin	III. Corn borer
D. alpha-1 antitrypsin	IV. Milk

Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-IV, D-I
- (2) A-III, B-II, C-IV, D-I
- (3) A-IV, B-II, C-I, D-III
- (4) A-II, B-IV, C-I, D-III

192. Select the **correct** statement(s).

- A. First menstrual cycle begins at puberty is called menopause.
- B. Lack of menstruation is indicative of pregnancy alone.
- C. In the absence of fertilization, the corpus luteum degenerates that causes disintegration of endometrium.
- D. Graafian follicles secrete the hormone, relaxin, during the later phase of pregnancy.

- (1) A and C only
- (2) C and D only
- (3) C only
- (4) D only

193. Match List-I with List-II:

List-I	List-II
A. Reptilia	I. Creep or crawl
B. Osteichthyes	II. Crop and gizzard
C. Hemichordates	III. Stomochord
D. Aves	IV. Air bladder

Choose the **correct** answer from the options given below:

- (1) A-III, B-I, C-IV, D-II
- (2) A-II, B-III, C-I, D-IV
- (3) A-I, B-IV, C-III, D-II
- (4) A-III, B-IV, C-II D-I

194. Given below are two statements:

Statement-I: Cell-mediated immune response is responsible for graft rejection.

Statement-II: Tissue matching is required before undertaking any graft.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement-I is correct but Statement-II is incorrect.
- (2) Statement-I is incorrect but Statement-II is correct.
- (3) Both Statement-I and Statement-II are correct.
- (4) Both Statement-I and Statement-II are incorrect.

195. Match List-I with List-II:

List-I	List-II
A. Glycosuria	I. Presence of glucose in urine
B. Ketonuria	II. Insoluble mass of crystallised salts
C. Renal calculi	III. Inflammation of glomeruli
D. Glomerulonephritis	IV. Presence of ketone bodies in urine

Choose the **correct** answer from the options given below:

- (1) A-IV, B-III, C-I, D-II
- (2) A-I, B-II, C-IV, D-III
- (3) A-II, B-III, C-IV, D-I
- (4) A-I, B-IV, C-II, D-III

196. Which of the following statements are **true** with respect to cockroach?

- A. Cockroaches are dioecious.
- B. Male reproductive system consists of a pair of testes, one lying on each lateral side in the 4th-6th abdominal segments.
- C. Each ovary is formed of a group of eight ovarioles.
- D. A pair of spermatheca is present in the 9th segment, which opens into the genital chamber.

Choose the *most appropriate* answer from the options given below:

- (1) A, B and C only
- (2) B, C and D only
- (3) C and D only
- (4) A, B and D only

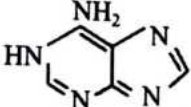
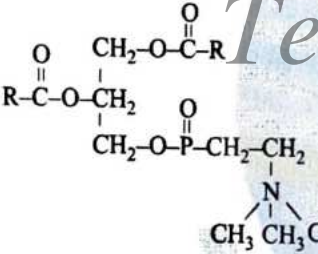
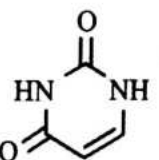
197. Match List-I with List-II:

List-I	List-II
A. Unipolar neurons	I. Transmit impulses from tissue/organ to central nervous system
B. Nodes of Ranvier	II. Transmit impulses from central nervous system to tissue/organ
C. Afferent nerve fibres	III. Cell body with one axon
D. Efferent nerve fibres	IV. Gaps between two adjacent myelin sheaths

Choose the **correct** answer from the options given below:

- (1) A-III, B-IV, C-I, D-II
- (2) A-I, B-II, C-IV, D-III
- (3) A-IV, B-III, C-II, D-I
- (4) A-II, B-IV, C-I, D-III

198. Match List-I with List-II:

List-I	List-II
A. $\begin{array}{c} \text{CH}_2\text{—OH} \\ \\ \text{CH—OH} \\ \\ \text{CH}_2\text{—OH} \end{array}$	I. Uracil
B. 	II. Adenine
C. 	III. Lecithin
D. 	IV. Glycerol

Choose the **correct** answer from the options given below:

- (1) A-IV, B-II, C-III, D-I
- (2) A-I, B-II, C-IV, D-III
- (3) A-II, B-III, C-IV, D-I
- (4) A-IV, B-III, C-II, D-I

199. Read the following statements:

- A. Scapula is situated in the dorsal part of the thorax between the second and the seventh ribs.
- B. The striated appearance of a myofibril is due to the distribution pattern of sarcoplasmic reticulum.
- C. The central hollow portion of the vertebra is called the neural canal.
- D. I-band shortens during muscle contraction.

How many of the above statement(s) is/are **incorrect**?

- (1) Two
- (2) Three
- (3) Four
- (4) One

200. Match List-I with List-II:

List-I	List-II
A. Isolation of genetic material in bacteria	I. Gene gun
B. Isolation of genetic material in plant cells	II. Spooling
C. Method of removing DNA that separates out on glass rod	III. Cellulase
D. Use micro-particles of gold or tungsten	IV. Lysozyme

Choose the **correct** answer from the options given below:

- (1) A-I, B-III, C-II, D-IV
- (2) A-II, B-III, C-IV, D-I
- (3) A-IV, B-II, C-I, D-III
- (4) A-IV, B-III, C-II, D-I